

FERDUS HOSSAIN

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate of American International University-Bangladesh (2026), CGPA 3.91/4.00, with Dean's List recognition in six consecutive semesters. Nine peer-reviewed conference papers published or accepted during undergraduate study, six as first author and seven as corresponding author, with five further manuscripts awaiting journal submission. Experienced in literature review, experimental design, statistical analysis, manuscript writing and conference presentation.

CORE COMPETENCIES

Data Collection • Statistical Analysis • Experimental Design • Literature Review • Machine Learning • Scientific Writing • Presentation • Research Ethics and Data Integrity

RESEARCH INTERESTS

Decision-making under uncertainty (CVaR and distributionally robust optimisation for UAV route planning); trustworthy machine learning (differential privacy, conformal prediction); applied statistics and survey-based analysis.

EDUCATION

American International University-Bangladesh	2026
B.Sc. in Computer Science and Engineering, Major in Information Systems CGPA: 3.91 / 4.00	
Dr. Mahbubur Rahman Mollah College	2021
HSC, Science GPA: 5.00 / 5.00	
Ideal School and College, Motijheel, Dhaka	2019
SSC, Science GPA: 4.94 / 5.00	

ACADEMIC HONORS

- Dean's List Award, AIUB | six consecutive semesters, Fall 2023–2024 through Spring 2025–2026.
- Magna Cum Laude, American International University-Bangladesh | anticipated at the 2028 convocation.

RESEARCH EXPERIENCE

Research Assistant, UCHRG Lab	May 2026 – Present
Ubiquitous, Cloud and HCI Research Group Supervisor: Prof. Dr. Kamruddin Md. Nur	
AIUB, Dhaka	
- Support two to three concurrent projects against fixed external deadlines, carrying each from literature review and experimental design through data analysis, internal review and final manuscript. - Design and run reproducible experiments in Python, including pre-registered analysis plans and multi-seed evaluation, and keep version-controlled records so every reported number can be traced to the run that produced it, upholding research ethics and data integrity. - Serve as corresponding author for IEEE and internationally indexed venues, handling submission, reviewer correspondence and camera-ready preparation, and present first-authored work at international conferences.	
Research Intern, Applied Intelligence and Informatics Lab	Jun 2026 – Sep 2026
Supervisor: Dr. Md. Saef Ullah Miah	
Remote, Nottingham, UK	
- Worked in a fully remote, cross-border team, setting weekly research priorities and reporting progress on a fixed weekly cycle. - Owned data collection, cleaning, validation and statistical analysis, and wrote up findings for shared research outputs.	
Departmental Intern, Department of Computer Science	2026 – Present
American International University-Bangladesh	
Dhaka	
First point of contact for undergraduate students on academic procedures and faculty availability in a high-volume, walk-in setting; designed and deployed the internal system that supports the service.	

PEER-REVIEWED PUBLICATIONS

Nine conference papers published or accepted; first author on six and corresponding author on seven.

- Risk-Aware Drone Routing with Stochastic Wind Fields
IEEE ICCIT 2025 (IEEE Xplore, Scopus) | First and corresponding author | Published, Dec 2025
- Bayesian Wind Forecast Correction for Real-Time CVaR-Based UAV Replanning
IEEE QPAIN 2026 (IEEE Xplore, Scopus) | First and corresponding author | Published, Apr 2026

3. **Vertical Wind-Profile Driven 3D UAV Routing with CVaR-Based Altitude Optimization**
IEEE QPAIN 2026 (IEEE Xplore, Scopus) | First and corresponding author | **Published, Apr 2026**
4. **Disease-Specific vs. Cross-Disease Clusters in PubMed: A Six-Embedding Benchmark at Scale**
ICCA 2026 | First and corresponding author | **Accepted, Aug 2026**
5. **DP-SGD Noise Direction Does Not Shape the Last-Layer Laplace Posterior: A Multi-Dataset Mechanistic Analysis**
ICCA 2026 | First and corresponding author | **Accepted, Aug 2026**
6. **Environmental Challenges of Textile Wastewater in Bangladesh**
ICTSE 2025, BUTEX International Conference | First author | **Published, May 2025**
7. **Determinants of Pay Satisfaction and Working Conditions Among Bangladeshi RMG Workers**
ICTSE 2026, BUTEX International Conference | Second author, corresponding author | **Published, Jun 2026**
8. **Incentive Pay, Style Disruptions and Overtime: Statistical Determinants of Productivity in Bangladeshi RMG**
ICTSE 2026, BUTEX International Conference | Second author | **Published, Jun 2026**
9. **Intelligent Waste Classification for Urban Bangladesh: A MobileNetV3 Approach**
PECCII 2026, Pabna University of Science and Technology | Second author, corresponding author | **Published, Jun 2026**

MANUSCRIPTS IN FINAL PREPARATION

Completed manuscripts under supervisor review prior to journal submission; first author on all five.

1. **Wasserstein Distributionally Robust Multi-Objective UAV Routing.**
2. **Physics-Constrained Differentially Private Federated Learning for Byzantine-Resilient Wind Hazard Estimation Across UAV Fleets.**
3. **Conformal Pareto Certificates: Distribution-Free Finite-Sample Guarantees for Multi-Objective UAV Routing Under Weather Uncertainty.**
4. **The Group Coverage Gap Is Mostly Noise: A Pre-Registered Negative Study of Score Design in Group-Conditional Conformal Prediction.**
5. **Limits of Bootstrap-Based Uncertainty Decomposition for Fair Conformal Prediction: An Empirical Study.**

RESEARCH SYSTEMS AND PROJECTS

- **TRACE** 2026 – Present
 Research tooling for large-scale scientific literature analysis, under development at the AIUB AI Supercomputer Lab. Covered by a non-disclosure agreement and not publicly listed; further detail available on request within its terms.
- **LegalMind: Local-Inference Document Intelligence Pipeline** May 2026
 End-to-end pipeline running entirely on local hardware, combining OCR, structured extraction, hybrid retrieval and claim-level verification with no external API dependency.

TUTORING AND TEACHING INTERESTS

Prepared to tutor undergraduate students weekly in Programming Fundamentals (Python), Object-Oriented Programming, and Data Structures and Algorithms.

TECHNICAL SKILLS

- **Programming and Tools:** Python (primary), C++; Git, Docker, LaTeX, llama.cpp, Zotero, EndNote.
- **Machine Learning:** PyTorch, scikit-learn; CNNs, vision transformers, transfer learning; DP-SGD with Rényi DP accounting, conformal prediction, CVaR and distributionally robust optimisation.
- **Statistics and Data Analysis:** pandas, NumPy, statsmodels, SPSS, R; survey design, exploratory factor analysis, ANOVA, regression, bootstrap confidence intervals.
- **Languages:** Bangla (native); English (fluent), including scientific writing for IEEE and Elsevier venues.

REFERENCES

- **Prof. Dr. Kamruddin Md. Nur** | Professor, Department of Computer Science, AIUB | kamruddin@aiub.edu
- **Dr. Md. Saef Ullah Miah** | Associate Professor and Director, Department of Computer Science, AIUB | saef@aiub.edu
- **Dr. Jannatul Fardus** | Assistant Professor, Department of Mathematics, AIUB | jannatul@aiub.edu